



1. IDENTIFICATION OF THE SUBSTANCE / PREPARATION AND OF THE COMPANY

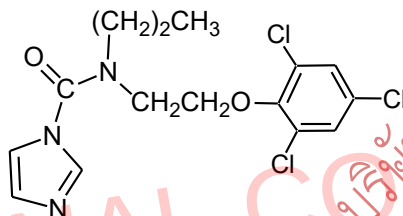
1.1 Identification:

Common name: Prochloraz 45% W/V EC

Chemical name: *N*-propyl-*N*-[2-(2,4,6-trichlorophenoxy)ethyl]imidazole-1-carboxamide

Empirical formula: C₁₅H₁₆Cl₃N₃O₂

Structural formula:



Molecular weight: 376.7

1.2 Uses: Fungicide

1.3 Manufacturer:

Manufacturer: The Erawan Agricultural Chemicals Co., Ltd

Address: 15/7 Soi 33 Ladprao Road. Chandahasam Chatuchak, Bangkok 10900 Thailand

Tel: 02 5135251

Fax: 02 5133181

2. HAZARDS IDENTIFICATION

GHS Classification of product

Flammable liquids – Category 3

Aspiration Hazard – Category 1

Acute Toxicity Oral – Category 4

Acute Toxicity Dermal – Category 5

Acute Inhalation – Category 4

Eye Damage – Category 1

Skin Irritation – Category 2

Specific Target Organ Toxicity Repeated Exposure – Category 2

Acute Aquatic Toxicity – Category 1

Chronic Aquatic Toxicity – Category 1

Label Elements

Classification and Labelling of Chemicals (GHS)

Rev 9, 2021; Regulation EC No. 1272/2008

[EU-GHS/CLP]



Signal word

DANGER

Hazard Statements

H226 – Flammable liquid and vapour

H302 – Harmful if swallowed

H304 – May be fatal if swallowed and enters airways

H313 – May be harmful in contact with skin

H332 – Harmful if inhaled

H315 – Causes skin irritation

H318 – Causes serious eye damage

H373 – May cause damage to organs through prolonged or repeated exposure

H410 – Very toxic to aquatic life with long lasting effects

General Precautionary Statements

P101: If medical advice is needed, have product container or label at hand.

P102: Keep out of reach of children.

P103: Read carefully and follow all instructions.

Prevention Precautionary Statements

P210: Keep away from heat, hot surfaces, sparks, open flames, and other ignition sources. No smoking.

P233: Keep container tightly closed.

P240: Ground and bond container and receiving equipment.

P241: Use explosion-proof [electrical, ventilating, lighting] equipment.

P242: Use non-sparking tools.

P243: Take action to prevent static discharges.

P260: Do not breathe mist/vapours/spray.

P264 + P265: Wash hands and face thoroughly after handling. Do not touch eyes.

P270: Do not eat, drink, or smoke when using this product.

P271: Use only outdoors or in a well-ventilated area.

P273: Avoid release to the environment.

P280: Wear protective gloves/protective clothing/eye and face protection.

Response Precautionary Statements

P301 + P316 + P331: IF SWALLOWED: Get emergency medical help immediately. Do NOT induce vomiting.

P303 + P361 + P353: IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse affected areas with soap and water or take a shower.

P332 + P317: If skin irritation occurs: Get medical help.

P363: Wash contaminated clothing before reuse.

P304 + P340 + P317: IF INHALED: Remove person to fresh air and keep comfortable for breathing. Get medical help.

P305 + P354 + P338 + P317: IF IN EYES: Immediately rinse with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical help.

P319: Get medical help if you feel unwell.

P321: For specific treatment, see first aid measures on this label.

P370 + P378: In case of fire: Use water spray, alcohol-resistant foam, dry chemical, or carbon dioxide CO₂ to extinguish the fire.

P391: Collect spillage.

Storage Precautionary Statements

P403 + P235: Store in a well-ventilated place. Keep cool.

P405: Store locked up.

Disposal Precautionary Statements

P501: Dispose of contents and container in accordance with national regulations.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Components	CAS No.	Percentage (% W/V)
Prochloraz	67747-09-5	45
Inert ingredient	-	Up to 100%

4. FIRST AID MEASURE

INHALATION:

- Remove the victim from immediate source of exposure. Move victim to fresh air, if it can be done safely, and keep comfortable.
- If victim's breathing has stopped, perform artificial respiration.
- DO NOT perform mouth-to-mouth resuscitation if victim ingested or inhaled the substance; wash face and mouth before giving artificial respiration. Use a pocket mask equipped with a one-way valve or other proper respiratory medical device.
- Administer oxygen if victim's breathing is difficult or irregular.
- Get medical help.

SKIN:

- Remove and isolate contaminated clothing, shoes, and leather goods immediately and take a shower.
- Rinse affected areas (skin) immediately with non-abrasive soap or mild detergent and large amounts of running water. Wash contaminated clothing before reuse.

- Get medical help if irritation develops and persists.
- In case of burns, IMMEDIATELY cool affected skin for as long as possible with cold water.

EYES:

- Rinse eyes IMMEDIATELY with clean running water for at least 15 minutes, while holding eyelids apart. Remove contact lenses after 5 minutes if present and easy to do. Continue rinsing while holding eyelids apart.
- Seek medical help if irritation continues.

INGESTION:

- If swallowed, DO NOT induce vomiting, unless instructed to do so by poison control center or doctor.
- Have person sip a glass of water if able to swallow.
- Never give anything by mouth to an unconscious person.
- If vomiting does occur, keep on giving fluids. Get medical help.

NOTE TO PHYSICIAN:

There is no specific antidote. All treatments should be based on observed signs and symptoms of distress in the patient

POTENTIAL HEALTH EFFECTS, ACUTE AND DELAYED

Effects of exposure (inhalation, ingestion, or skin contact) to substance may be delayed.

5. FIRE FIGHTING MEASURE**SUITABLE EXTINGUISHING MEDIA:**

Small fires: Dry chemical powder, carbon dioxide (CO₂), water spray or alcohol-resistant foam

Large fires: Water spray, fog, alcohol-resistant foam, or carbon dioxide (CO₂)

FIRE INVOLVING TANKS:

Cool containers with flooding quantities of water until well after fire is out. DO NOT get water inside containers. Withdraw immediately in case of rising sound from venting safety devices or discoloration of tank. Always stay away from tanks engulfed in fire.

UNSUITABLE EXTINGUISHING MEDIA:

DO NOT use high volume water jet, due to contamination risk.

SPECIFIC EXTINGUISHING METHODS:

Fight fire from maximum distance. For massive fire, use unmanned hose holder or monitor nozzles. Collect contaminated extinguishing water separately; do not allow contaminated water to reach the sewage or effluent systems.

SPECIFIC HAZARDS ARISING FROM COMBUSTION

In case of fire, the formation of Carbon monoxide (CO), carbon dioxide (CO₂),

PRODUCTS: Ammonia, and Hydrogen chloride (HCl) can be expected.

PRECAUTIONS FOR FIRE FIGHTERS:

Fire fighters should wear full protective gear including self-contained breathing apparatus (SCBA). Fight fire from safe distance. Contact with the fumes and vapours should be avoided by staying upwind. Clean all clothing before reuse. Severely contaminated clothing cannot be adequately decontaminated and must be disposed as a hazardous waste. Shower with soap and water after contact with chemical product.

FURTHER INFORMATION:

- If possible, safely move undamaged intact containers away from the area around the fire.
- Keep containers cool by spraying with water if exposed to fire.
- Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.
- In case of fire and/or explosion do not breathe fumes.

6. ACCIDENTAL RELEASE MEASURES

PERSONAL PRECAUTIONS:

Avoid contact with skin and eyes. Do not touch or walk through spilled material. Do not inhale spray or fumes.

PROTECTIVE EQUIPMENT:

Wear personal protective clothing and equipment (see section 8).

EMERGENCY PROCEDURES:

Keep people and animals away. Eliminate all ignition sources (no smoking, flares, sparks, or flames) from immediate area. All equipment used when handling the product must be grounded.

Use water spray to reduce vapours or divert vapour cloud drift.

ENVIRONMENTAL PRECAUTIONS:

PREVENT spilled material from entering waterway and sewer systems, basements, and confined areas. If the product contaminates rivers and lakes or waterways immediately inform respective authorities.

METHODS AND MATERIALS FOR CONTAINMENT:

Contain and absorb liquid spills with inert material, remove by scoop or vacuum. Use approved industrial vacuum cleaner for removal and place in clearly marked waste containers.

METHODS AND MATERIALS FOR CLEANING UP:

Contain spillage, and then collect with non-combustible absorbent material, (e.g., sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Use clean, non-sparking tools to collect absorbed material.

7. HANDLING AND STORAGE**7.1 PRECAUTIONS FOR SAFE HANDLING:**

- Always store fungicides in their original containers, which include the label listing ingredients, directions for use, and first aid steps in case of accidental poisoning.
- Never transfer fungicides to soft drink bottles or other containers. Children or others may mistake them for something to eat or drink.
- Wear suitable protective clothing which include chemical-resistant overalls, footwear, socks, dust mask, eye shields and gloves.
- Wash hands before eating, drinking, chewing gum, smoking, or using the toilet. Wash hands, arms, and face after application. Wash gloves and contaminated protective clothing daily before reuse.

7.2: CONDITIONS FOR SAFE STORAGE:

- Keep out of reach of unauthorized persons, children, and animals. Always store fungicides in their original containers, closed with original cap and the original label, in a cool, dry, and well-ventilated area out of direct sunlight.
- Segregate from foods and animal feeds.
- DO NOT reuse the container for any other purpose.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

ADI – Acceptable Daily Intake 0.01 mg kg⁻¹ bw day-1

AOEL – Accepted Operator Exposure Level 0.02 mg kg⁻¹ bw day-1

ENGINEERING CONTROLS: Containment and/or segregation is the most reliable technical protection measure if exposure cannot be eliminated. If airborne mist/vapours are generated use local exhaust ventilation controls. Facilities should be equipped with an eyewash station and a safety shower. Where necessary, seek additional occupational hygiene advice.

PERSONAL PROTECTIVE EQUIPMENT:

Respiratory Protection: Where exposure through inhalation may occur when handling and/or when preparing the spray mixture, wear a face mask. If the product is used in confined spaces a respirator suitable for protection from dusts and mists of pesticides is adequate.

Hand Protection: Wear chemical-resistant gloves made of any waterproof material such as nitrile rubber. Glove thickness: 0.5 mm

Eye Protection: The use of safety goggles (full-face shield) is recommended.

Skin and Body Protection: Wear suitable protective clothing which include chemical-resistant overalls, footwear, socks, dust mask, eye shields and gloves. Remove and wash contaminated protective clothing daily.

9. PHYSICAL AND CHEMICAL PROPERTIES

PHYSICAL STATE: Liquid

COLOUR: Clear, Light yellow to brown

ODOUR: Slight organic odour (aromatic)

MELTING POINT / FREEZING POINT °C: 48 °C (a.i prochloraz)

BOILING POINT: Decompose before boiling

DECOMPOSITION TEMPERATURE (a.i): 220 °C (a.i prochloraz)

FLAMMABILITY: Flammable liquid – Category 3

FLASH POINT: 59 °C

pH (1% IN WATER): 6.0 – 9.0

DENSITY / RELATIVE DENSITY: 1.42 g/ml

SOLUBILITY - WATER (a.i): 26.5 mg/l (a.i prochloraz)

N-OCTANOL / WATER PARTITION COEFFICIENT (a.i): log Pow = 3.5 at 20 °C (a.i prochloraz)

VAPOUR PRESSURE: 0.15 mPa at 20 °C (a.i prochloraz)

10. STABILITY AND REACTIVITY

REACTIVITY: Stable under normal conditions no reaction with fire-fighting water.

CHEMICAL STABILITY: Stable under normal use and storage conditions for at least 2 years.

HAZARDOUS REACTION: Hazardous polymerization is not expected to occur.

CONDITIONS TO AVOID: Avoid storage in moist or hot conditions, near to heat or ignition sources.

(e.g. – heat, pressure, static discharge, shock, or vibration)

Keep away from food, drink, and open bodies of water.

INCOMPATIBLE MATERIALS: None specified, Strong acids, Strong oxidizers, and Strong bases.

HAZARDOUS DECOMPOSITION PRODUCTS: When heated to decomposition, irritant or dangerous fumes/vapours may be emitted. See section 5.

11. TOXICOLOGICAL INFORMATION

ORAL:	LD ₅₀ (rat) = 1669.74 mg a.i. /kg bw	Category 4
DERMAL:	LD ₅₀ (rat) = 3243.74 mg a.i. /kg bw	Category 5
INHALATION:	LC ₅₀ (4h) rat = 3.62 mg/l	Category 4
SKIN IRRITATION / CORROSION:		Category 2
SERIOUS EYE IRRITATION / DAMAGE:		Category 1
RESPIRATORY OR SKIN SENSITIZATION:		Not Classified
GERM CELL MUTAGENICITY:		Not Classified
CARCINOGENICITY:		Not Classified
REPRODUCTIVE TOXICITY:		Not Classified
SPECIFIC TARGET ORGAN TOXICITY – SINGLE EXPOSURE:		Not Classified
SPECIFIC TARGET ORGAN TOXICITY – REPEATED EXPOSURE:		Category 2
ASPIRATION HAZARD:		Category 1

12. ECOLOGICAL INFORMATION

PROCHLORAZ	Birds: LD ₅₀ (oral)	Colinus virginianus (Bobwhite quail) Acute LD ₅₀ > 662 mg a.i./kg bw/day LC ₅₀ /LD ₅₀ > 1580 mg a.i. /kg diet Chronic NOEL = 33.1 mg a.i. /kg bw/day
	Fish: LC ₅₀	Oncorhynchus mykiss (Rainbow trout) Acute (96h) LC ₅₀ = 1.5 mg a.i./l <i>Pimephales promelas</i> (Fathead minnow) NOEC = 0.049 mg a.i./l (36-day)
	Aquatic invertebrates	Daphnia magna (Water flea) Acute (48h) EC ₅₀ = 4.3 mg a.i./l Chronic (21-day) NOEC = 0.022 mg a.i./l
	Aquatic crustaceans	Americamysis bahia Acute (96h) LC ₅₀ = 0.77 mg a.i./l
	Algae - EC ₅₀ / NOEC	Scenedemus subcapitata Acute (72h) EC ₅₀ > 0.0055 mg a.i./l Chronic (96h) NOEC = 0.01 mg a.i./l
	Bees	Apis mellifera Acute contact 48-hour LD ₅₀ > 141.3 (µg bee-1) Acute oral 48-hour LD ₅₀ > 101 (µg bee-1) Not toxic to bees
	Earthworms: LC ₅₀ /NOEC	Eisenia fetida Acute (14-day) LC ₅₀ > 500 mg a.i./kg d.w. soil Chronic NOEC = 4.2 mg a.i./kg d.w.

PERSISTENCE, DEGRADABILITY AND MOBILITY:

Prochloraz is moderately to persistent in the soil and slightly mobile.

Prochloraz	DT ₅₀ = 69 – 120 days Koc = 500 (slightly mobile)
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13. DISPOSAL CONSIDERATIONS

On site disposal of the concentrated product is not acceptable. Ideally, the product should be used for its intended purpose. If there is a need to dispose of the product, approach local authorities.

TRIPLE RINSE THE EMPTY CONTAINER AS FOLLOWS: Containers must be completely emptied before being disposed of. Invert the empty container over the spray or mixing tank and drain for at least 30 seconds until the flow has slowed down to a drip. Thereafter rinse the empty container three times in succession with one quarter of the container volume fresh water and decant the rinsate into the spray or mixing tank. Puncture the triple rinsed container and dispose of via an approved collector or recycler. Do not bury, burn, or donate the container to any other parties that may use it as a container for food or beverages.

14. TRANSPORT INFORMATION

UN NUMBER: 1993

UN PROPER SHIPPING NAME: FLAMMABLE LIQUID, (45% Prochloraz)
ENVIRONMENTALLY HAZARDOUS SUBSTANCES, LIQUID, Prochloraz)

TRANSPORT HAZARD CLASS(ES): Class 3 (Subsidiary hazard 9)

PACKING GROUP: III (low danger)

TRANSPORT PICTOGRAMS:



ENVIRONMENTAL HAZARD: Marine Pollutant: Yes – Category 1

TRANSPORT IN BULK: Not applicable, not to be transported in bulk.

SPECIAL PRECAUTIONS FOR USER: Not applicable

FLAMMABLE LIQUID, (45% Prochloraz)

ENVIRONMENTALLY HAZARDOUS SUBSTANCES, LIQUID, (45% Prochloraz and 30 – 50% petroleum solvent)

15. REGULATORY INFORMATION

Code letter and hazard designation of product: Not available

Hazard-determining components of labeling: Not available

Risk phrases: may be irritating to eyes, skin and respiratory system

Safety phrases: in case of accident, flush eyes or skin with copious amounts of water; remove contaminated clothing and shoes; call a physician

16. OTHER INFORMATION

The information above is believed to be accurate and represents the best information currently available to us. However, we make no warranty of merchantability or any other warranty, express or implied, with respect to such information, and we assume no liability resulting from its use. Users should make their own investigations to determine the suitability of the information for their particular purposes. In no event shall we be liable for any claims, losses, or damages of any third party or for lost profits or any special, indirect, incidental, consequential or exemplary damages, howsoever arising, even if we have been advised of the possibility of such damages.